

Hariton Pitsik

JUNIOR ML ENGINEER · ML INTERN

Saratov, Russia

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Education

Saratov State University

BACHELOR | FUNDAMENTAL COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Saratov, Russia

September 2023 - June 2027

Skills

Programming languages

Python, Rust, SQL

Spoken languages

Russian (native), English (B2)

Technologies

Git & GitHub, Docker, Linux, LaTeX, Typst, Jupyter, NumPy, Pandas, Matplotlib & Seaborn, Scikit-learn, XGBoost, CatBoost, FastAPI, ClearML, Streamlit, PyTorch, Transformers, PostgreSQL

Concepts

Mathematics & Statistics (Linear Algebra, Calculus, Probability Theory), Classical ML (Supervised & Unsupervised Learning), Deep Learning, Natural Language Processing & LLM, Computer Vision

Some achievements

Teaching & Social

- Participant in Development Students Clubs (DSC), a student-led community that improves faculty life ([DSC website](#)).
- Teach machine learning as part of the DSC community.
 - Lecture recordings on NumPy ([YouTube](#)) and Pandas ([YouTube](#)).
 - Machine Learning Club playlist: ([YouTube](#))

Practical Experience

Speech-to-Text System for Dialogues and Catalog Terms

🌐 haritonn/asr_practice

PRODUCTION INTERNSHIP PROJECT.

- Developed a speech-to-text system for Russian-language dialogues, combining Faster-Whisper transcription, Silero VAD, and pyannote speaker diarization;
- Implemented recognition of domain-specific catalog terms with a NeMo CTC context graph, eliminating the need to fine-tune the ASR model;
- Linked transcripts and detected terms to dialogue participants; evaluated solution quality using WER, CER, DER, and terminology F1.

Projects

Local RAG System

🌐 haritonn/pdf_rag

- Built a fully local RAG assistant that answers questions over uploaded PDF documents;
- Made retrieval configurable from the UI: switchable LLM, embedding model, vector database, and Top-K sources;
- Made answers auditable by exposing source documents and retrieved contexts in the Streamlit interface; implemented a modular architecture with Qdrant.

Biomedical Search

 [haritonn/fitness_se](#)

- Built a search engine over 23,283 Europe PMC papers on training, recovery, and sports nutrition using weighted BM25, MedCPT with FAISS, and hybrid retrieval with Reciprocal Rank Fusion;
- Implemented offline evaluation on an internal set of 150 queries and 3,000 graded relevance judgments using nDCG, MAP, Precision, Recall, and Judged@K; BM25 reached test nDCG@10 of 0.5019;
- Built a local Ollama/Qwen3 8B-assisted relevance-labeling pipeline pooling BM25, dense, and hybrid candidates; added a Streamlit interface and GitHub Actions CI.

Caption Generator

 [haritonn/caption_gen](#)

- Engineered an end-to-end image-captioning pipeline on Flickr8k: ResNet-50 encoder + soft-attention LSTM decoder in PyTorch;
- Applied scheduled sampling, label smoothing, attention regularization, gradient clipping, and early stopping to improve training stability;
- Built a reproducible experiment workflow with deterministic train/validation/test splits, checkpointing, and optional ClearML tracking;
- Evaluated caption quality with BLEU and METEOR; reached 0.1655 BLEU and 0.3774 METEOR on the validation split (37m. weights).

cargo-smi — NVIDIA GPU Monitor

 [haritonn/cargo-smi](#)

- Developed a responsive terminal dashboard in Rust for unified monitoring of NVIDIA GPUs and host-system resources;
- Integrated NVML to display per-GPU temperature, utilization, VRAM, CUDA/driver versions, and GPU-consuming processes;
- Added multi-GPU navigation, configurable auto-refresh, manual refresh, and a 120-sample utilization history chart;
- Surfaced CPU, RAM, swap, and the 20 most CPU-intensive processes in the same TUI using Ratatui, Crossterm, and Sysinfo.

ResNet with Attention

 [haritonn/resnet_attention](#)

- Implemented baseline ResNet-50 and a channel-attention variant from scratch in PyTorch;
- Improved classification quality over the baseline by adding channel attention; compared convergence, accuracy, and F1 score;
- Built a reproducible training and inference workflow with early stopping, attention visualizations, and ClearML experiment tracking.

Coursework

2nd Year: Generative Computer Vision

 [haritonn/coursework2](#)

COMPARED DIFFUSION MODELS, GANS, AND LARGE MULTIMODAL MODEL (LMM)-BASED APPROACHES FOR VIRTUAL TRY-ON. EVALUATED VISUAL QUALITY AND INFERENCE TIME ON A CUSTOM DATASET.

3rd Year: Text-Embedding Geometry & HDBSCAN Clustering

 [haritonn/coursework3](#)

INVESTIGATED HOW EMBEDDING GEOMETRY AFFECTS NEWS-EVENT CLUSTERING: RAN 2,604 HDBSCAN EXPERIMENTS ACROSS 3 EMBEDDING FAMILIES, 93 FEATURE SPACES, AND 28 PARAMETER VALUES. SHOWED THAT UMAP IMPROVED THE BEST F1 FOR EVERY EMBEDDING FAMILY; ACHIEVED $F1 = 0.837$ WITH JINA EMBEDDINGS V5 + UMAP(10).